

1- A nozzle is designed to expand a gas with $\gamma=1.2$ and $M=30$ g/mol from a chamber where the pressure and temperature are 1000 KPa and 800 C, respectively, to a Mach number of 4. The throat area of the nozzle is 0.02 m^2 . Find:

- The exit area of the nozzle.
- The mass flow rate through the nozzle when operating at design conditions.
- The back pressure for which it will be a normal shock wave on the exit plane of the nozzle.
- The range of back pressures where there will be a normal shock wave inside the nozzle.
- The range of back pressures over which oblique shocks occur outside the nozzle.
- The range of pressures over which expansion waves will be outside the nozzle.
- Find the altitude corresponding to the pressures of cases c-f. Write a MATLAB code that gets pressure and gives altitude.

2- Sea level air ($P=100$ Kpa, $T=288$ K) is being drawn isentropically through a C-D nozzle into an exit tank. The cross-sectional areas of the duct at the mouth, throat and entrance to the vacuum tank are 2 m^2 , 1 m^2 , and 4 m^2 , respectively. Find:

- Maximum air that can be drawn into the vacuum tank (Hint: Throat chocks)
- If the maximum flow rate is attained, find the pressure (P_e) in the exit tank
- If a normal shock is detected (by Schlieren photography) in the divergence portion of the duct where the area is 3 m^2 , what will be the pressure (P_e) in the exit tank?

